

GENERAL SCIENCE — GRADE 10 | NUTRITION IN ANIMALS

Quick Check

Lesson 3: Enzymes and Regions of the Alimentary Canal — The Body's Chemical Toolkit

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.3: Nutrition in Animals | Lesson 3

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A research group fills in an enzyme chart for the mouth. Which row is correct for salivary amylase?

- A** Secreted by the stomach; acts on protein; produces peptides
- B** Secreted by the salivary glands; acts on starch (for example in ugali); produces maltose
- C** Secreted by the salivary glands; acts on starch; produces glucose only
- D** Secreted by the pancreas; acts on fats; produces fatty acids and glycerol

Lipase is an enzyme in the small intestine. What does lipase break down, and what does it produce?

- A** Peptides into amino acids
- B** Fats into fatty acids and glycerol
- C** Proteins into peptides
- D** Starch into maltose

A learner eats a piece of nyama choma (grilled meat), which is rich in protein. Which statement correctly describes what pepsin does to it?

- A** Pepsin is made in the pancreas and breaks fats into fatty acids and glycerol
- B** Pepsin is made by the gastric glands and breaks protein directly into glucose
- C** Pepsin is made by the gastric glands in the stomach and breaks protein into peptides
- D** Pepsin is made in the mouth and breaks starch into maltose

Why can salivary amylase in the mouth not break down the protein in nyama choma?

- A** Because salivary amylase is made in amounts that are too small to work on any food
- B** Because chewing destroys salivary amylase before it can act on protein
- C** Because the mouth is too warm for any enzyme to work on protein
- D** Because enzymes are specific: each enzyme acts only on its own substrate, and amylase acts on starch, not protein

The mouth and stomach each have only one or two main enzymes, but the small intestine has many. What is the best explanation?

- A** Every enzyme in the small intestine breaks down all food types, but each one works slowly
- B** The small intestine receives starch, proteins and fats, and each type of food needs its own enzyme
- C** Enzymes in the mouth and stomach are used up, so the small intestine must replace them
- D** The small intestine is narrow, so it needs more enzymes to push food along

When colour-coding an alimentary canal diagram, a learner uses one colour for protein-digesting enzymes. Which pair of enzymes should both be in that colour?

- A** Pepsin and trypsin
- B** Pepsin and salivary amylase
- C** Pancreatic amylase and pepsin
- D** Lipase and trypsin

In the small intestine, intestinal peptidases act on peptides that came from protein digestion. What is the final product of this step?

- A** Amino acids
- B** Starch
- C** Maltose
- D** Fatty acids and glycerol

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).